Quantified Analysis of Past Major League Baseball Data: An Exploration of Data in Python

(Milestone 1)

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CISD/CISP 357: Introduction to Data Science

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Section 1: Team Identification and Task Allocation Plan

The project team is interested in investigating the Major League Baseball ("MLB") historical data for trends and analysis to identify correlation between multiple historical datapoints. This information can be used for further exploration by the relevant subject matter experts.

The team believes the following project title aligns closely with the study's purpose:

Project Title: Quantified Analysis of Past Major League Baseball Data: An Exploration of Data in Python

An assignment agreement was created, identifying each member's responsibility to generate the identified deliverables. This does not mean each team member works in isolation, but only the tasks they are responsible for performing. Each team member is expected to reach out to the appropriate individuals as needed, either independently or as a team. See Table 1 for the agreed upon responsibility matrix.

Table 1

Assignment Responsibility Matrix

Student Name	Responsibility
Tristan Garica	• Collect appropriate data sources • Script Cleaning Code • Calculations Code • Report Reviewer
Benjamin Hays	• Managing GitHub (https://github.com/TechFluentBenjamin/CISD357_Final_Project) • Data Refining Proposal/ Requirements • Calculations Ideas/ flow • Report Writer
Both	• User Interface Design • Chart Design • Chart Code

Note. The following individuals can be contacted for information and consulting on an ad-hoc basis:

- "Jim" Python tutor at Sacramento City College
- Professor Fisher

Section 2: Topic Introduction and Domain Context

It is common knowledge that American baseball is America's pastime. In 2024, the MLB added approximately $10 billion dollars to the United States economy (Levin 2024). At the local level, MLB teams increase local property value, resulting in a higher demand for the surrounding property (Bradbury 2022). Clearly, as the MLB goes, so goes America.

Given the impact of MLB to the local economy, it is important for both teams and the local government to encourage good MLB governance. The data explored in this paper is vital for MLB teams to decide which players would thrive within the MLB system, the local government who implement fiscal policies to encourage MLB teams to relocate or stay within their jurisdiction, and potential investigations by boffins to identify the underlaying triggers identified in this investigation. Secondary groups impacted by this study includes city planners,

health scientists, and economics who may set financial policy or develop the relevant mathematical models.

Section 3: Data Source Specification & Metadata

The team propose to use the following sources for our data. These sources are from publicly available information which can be accessed as a .cvs file from the main website, https://sabr.org/lahman-database/. The specific databases for this project are identified in Table 2.

Table 2

Data Source Specifications and Metadata

Source Number	Metadata Property	Dataset Detail
1	Dataset Name	Fielding.csv from Lahman
	Publishing Organization	Lahman Baseball Database
	File Format	csv
	Approximate Row Count	~174,000
	Column Count/ Attributes	13
	Category Columns Used	PlayerID POS
2	Dataset Name	Batting.csv from Lahman
	Publishing Organization	Lahman Baseball Database
	File Format	.csv
	Approximate Row Count	~128,000
	Column Count/ Attributes	22 columns
	Category Columns Used	PlayerID Year RBI
3	Dataset Name	People.csv from Lahman
	Publishing Organization	Lahman Baseball Database
	File Format	.csv
	Approximate Row Count	~24,000
	Column Count/ Attributes	25
	Category Columns Used	playerID nameFirst nameLast weight height birthMonth debut finalGame
4.	Dataset Name	CollegePlaying.csv from Lahman
	Publishing Organization	Lahman Baseball Database
	File Format	.csv
	Approximate Row Count	~17,000
	Column Count/ Attributes	3
	Category Columns Used	playerID schoolID

Note. All information is potential and may change as knowledge of the assignment becomes clear.

Section 4: Research Questions & Hypotheses

Using the identified data sources, the team proposes four questions to investigate. The team understands only two questions are required for the assignment, however by offering multiple questions will allow the professor to narrow the team's scope, and will be a contingency if any of the proposed questions do not meet the assignment's criteria.

Because of data noise, the team expects a slight correlation caused by data randomness. This paper will use the following values for statistical conclusions:

0% to 5% = statically insignificant

>5% to 10% = weak correlation

>10% = strong correlation.

Research Question 1 (RQ1)

Research Question 1 (RQ1): Is there a connection between the salary of players to the number of runs batted in ("RBI") for 2024?

Hypothesis 1 (H1): The team believes there is a strong correlation between the salaries of players to the numbers of runs batted in for 2024. The reason is because scoring is very important to baseball, and the more RBIs a player can produce on average, the greater they should be paid.

Research Question 2 (RQ2)

Research Question 2 (RQ2): Has the statistical relationship between a player's height and their defensive position changed over the last 100 years?

Hypothesis 2 (H2): The team believes there is a strong likelihood that a player's height and the defensive position played has changed within the last 100 years. The variables causing this change is because of increased knowledge of general and sports medical care, increase

supply of food to young children, resulting in children growing up taller, and a realization by baseball managers that certain positions are played better by players of certain heights.

Research Question 3 (RQ3)

Research Question 3 (RQ3): Is there a connection between RBIs and player's birth month? In other words, does the month the player was born in influence the total amount of RBIs in their career? To minimize data noise, only retired players will be analyzed for this question.

Hypothesis 3 (H3): The team believes there is a weak correlation between RBIs and birth months for MLB players. There are multiple variables that affect a player's ability to hit RBIs, including the birth location, or high school location elevation. Additionally, a player's genetic makeup may influence oxygen saturation level (see, for example, Gibbens [2018] article on the Bajau's people ability to stay underwater for up to 13 minutes without physical harm).

Research Question 4 (RQ4)

Research Question 4 (RQ4): Is there a connection between the college type (i.e. Division 1 college, Division 2 college, etc.) the player attended, if any, and the number of seasons they play?

Hypothesis 4 (H4): There is a connection between the colleges they play and number of the seasons of a player. Different colleges impact different quality of baseball knowledge on potential baseball players, and therefore there is a connection between the college a player attends and the number of seasons they play.

Section 5: Small Data Subset Strategy

The team proposes the following steps to extract and analyze the data for completeness. The appropriate expected data will be identified for each column, such as date written in

month/date/year format, string, or integer. Multiple .cvs files or databases will be merged using a left union, with the “playerID” being the primary key connecting the files together, and any missing data will be replaced with the appropriate symbol or word for “missing data”. The first 100 rows of the merged document will be extracted using the relevant Python code. From that 100 rows, multiple checks will be performed on the data, such as confirming all rows meet the expected data format, data in missing rows automatically converted, and related actions performed.

A testing function will be developed using the Python unittest framework to confirm all extracted rows meet the required data expectations. This will provide feedback to the team to adjust the program code as needed.

Section 6: Calibrated Analytical Scope & Planned Visualizations

The team expects to use the following statistical metrics and visualization types for each research question. The team expects only two out of four of the following research questions to be approved. See Table 3 for details.

Table 3

Proposed Calibrated Analytical Scope & Planned Visualizations

Planned Descriptive Metrics	**Planned Visualizations (`matplotlib`)**	**Target Research Question**
Medium	Scatter plot	RQ1
Medium IQR	Faceted chart (X axis year, Y axis Medium Height, multiple charts)	RQ2
Medium Count Mean Standard deviation	Box plot	RQ3
Mean	Bar chart OR box plot depending on outcome	RQ4

Note. All information is potential and may change as knowledge of the assignment becomes clear.

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References

Bradbury, J.C. 2022. “Does Hosting a Professional Sports Team Benefit the Local Community? Evidence from Property Assessments.” *Economics of Governance* 22(3). doi:10.1007/s10101-022-00268-z.

Gibbens, S. 2018. “‘Sea Nomads’ Are First Known Humans Genetically Adapted to Diving.” *National Geographic* (5).

Levin, A. 2024. “Can Sports Drive Economic Development?” https://www.congress.gov/crs-product/IF12571.